

Interface Methods Usage Java



can I use interface methods without overriding in java



In Java, an interface is a contract that requires classes implementing it to override and provide concrete implementations for its methods (except for `default` methods, which already have a concrete implementation). If you attempt to use an interface method in a class without overriding it, it won't work unless one of the following applies:

1. Abstract Classes

If your class is declared as `abstract`, you can choose not to override the interface methods. This allows the subclass of your abstract class to override the methods instead.

```
java

interface MyInterface {
    void myMethod();
}

abstract class MyAbstractClass implements MyInterface {
    // Not overriding myMethod here is allowed because the class is abstract
}

class MyConcreteClass extends MyAbstractClass {
    @Override
    public void myMethod() {
        System.out.println("Implemented in concrete class.");
    }
}
```

2. `default` Methods in Interfaces

Starting with Java 8, interfaces can define `default` methods, which have an actual implementation. Classes implementing such an interface can use these methods without overriding them.

```
java

interface MyInterface {
    default void myMethod() {
        System.out.println("Default implementation.");
    }
}

class MyClass implements MyInterface {
    // No need to override myMethod, as the interface provides a default implementation
}

public class Main {
    public static void main(String[] args) {
```

```
MyClass obj = new MyClass();
obj.myMethod(); // Output: Default implementation.
}
}
```

3. Static Methods in Interfaces

Java interfaces can also contain `static` methods (since Java 8). These methods belong to the interface itself and are not meant to be overridden.

```
java

interface MyInterface {
    static void staticMethod() {
        System.out.println("This is a static method in the interface.");
    }
}

public class Main {
    public static void main(String[] args) {
        MyInterface.staticMethod(); // Call the static method directly using the interface name
    }
}
```

Summary

- **Required to override:** Interface methods without `default` implementations.
- **Not required to override:**
 - Methods in `abstract` classes that implement interfaces.
 - `default` methods.
 - `static` methods in interfaces (called directly via the interface name).

If you don't provide an override in a concrete class for methods without `default` implementations, you will get a compilation error.